Definition (Category of effectful procedures **Eff**)
Fix a set 🌍 of all possible worlds. The category **Eff** is defined by the following:

- *Objects*: same as the objects of $\langle\!\langle \mathbf{Set} \rangle\!\rangle$;
- *Morphisms*: a morphism $f : X \to_{\mathbf{Eff}} Y$ is a morphism

$$\mathsf{rep}(f) : \langle\!\langle X, 🌍 \rangle\!\rangle \to_{\langle\!\langle \mathbf{Set} \rangle\!\rangle} \langle\!\langle Y, 🌍 \rangle\!\rangle.$$

- *Composition*: The composition of $f : X \to_{\mathbf{Eff}} Y$ and $g : Y \to_{\mathbf{Eff}} Z$ is the morphism $f \fatsemi g : X \to_{\mathbf{Eff}} Z$ with $\mathsf{rep}(f \fatsemi g)$ given by

$$\mathsf{rep}(f \fatsemi_{\mathbf{Eff}} g) = \mathsf{rep}(f) \fatsemi_{\langle\!\langle \mathbf{Set} \rangle\!\rangle} \mathsf{rep}(g)$$

as illustrated in **??**.